

FarmGuard AI — Project Report

FutureTech HackFest 2026

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Theme: Innovating Tomorrow with Emerging Technologies

1. Introduction

FarmGuard AI is a full-stack smart agriculture web platform that uses AI and ML to help farmers detect crop diseases early and receive treatment recommendations. Designed for small farmers in India who lack access to agricultural experts.

2. Problem Statement

Small farmers lose 20–40% of crops annually due to late disease detection. Rural areas lack agricultural experts. Traditional diagnosis is slow, costly, and reactive.

3. Objectives

- Instant crop disease detection via photo upload
- AI confidence scores and severity classification
- Practical treatment recommendations
- Farmer dashboard with scan history
- No expensive hardware required

4. Proposed Solution

Web platform: Login → Select crop → Upload leaf photo → AI analysis → Disease + treatment → Dashboard tracking.

5. Technology Stack

Layer	Technology
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Frontend	Next.js 14, React, TypeScript, Tailwind CSS
Backend	Node.js, Next.js API Routes
AI/ML	Hugging Face Vision Transformer
Deploy	Vercel Cloud
Languages	TypeScript, JavaScript, HTML, CSS

6. Key Features

- Farmer authentication
- AI disease scanner with image upload
- Confidence score and severity levels
- Treatment recommendation engine
- Analytics dashboard with history
- Mobile-responsive design

7. Innovation

- Accessibility-first for rural farmers
- Instant results in seconds
- Zero hardware cost
- Full-stack AI integration
- Sustainable farming impact

8. Results

Working full-stack app with login, AI disease detection, treatment advice, and dashboard. GitHub: github.com/gowthamcodes225/farmguard-ai

9. Future Scope

- Mobile app (Android/iOS)
- IoT soil sensors
- Tamil/Hindi language support

- Disease outbreak heatmaps
- Government/NGO partnerships

10. Conclusion

FarmGuard AI shows how emerging AI technologies can solve real agricultural challenges and support sustainable farming in India.

Declaration: I declare this project is my original work for FutureTech HackFest 2026.